Year 3, Summer Semester (3 Credits)			
ELEC 499	Co-op Work Experience	3	ENGL 217
The above 118 credits complete the requirements for a Bachelor of Science degree in Electrical Engineering.			
Year 4, Fall Semester (16 Credits)			
CCEE 534	Signal Processing	3	CCEE 331
ELEC 534	Advanced Power Systems	3	ELEC 330
ELEC 595A	BE Summative Learning Project 1	1	ENGL 217, Senior standing
ENGR 510	Engineering Project Management	3	ENGR 300
	Engineering Technical Elective 2	3	
	Engineering Technical Elective 3	3	
Year 4, Spring Semester (18 Credits)			
ELEC 533	Advanced Machines and Drives	3	ELEC 331, 432
ELEC 563	Smart Grid Technology	3	ELEC 432
ELEC 595B	BE Summative Learning Project 2	3	ELEC 595A
	Engineering Technical Elective 4	3	
	Engineering Technical Elective 5	3	
	Social Sciences/ Humanities Elective II	3	
Total BE credits: 152			

* BS-bound students are required to take ELEC 400 – Summative Learning (BS) Project instead.

Note: Engineering Technical Electives (levels 400 or above) are selected as such:
6 credits restricted Departmental Electives
9 credits from any Engineering discipline at the 400 level or above.

Courses Description

I. Mandatory Courses

Non-Major Courses

Description of the mandatory, non-major courses follows.

CCEE 214	**Introduction to Programming**	**3(2,2)**
This course presents the fundamentals of structured programming concepts. It covers primitive data types, expressions, control statements, functions, arrays, basic search and sorting algorithms, and an introduction to pointers. **Prerequisite**: None. Equivalent to COSC 214.		

CCEE 221	**Logic Design**	**3(3,0)**
This course addresses Boolean algebra and logic simplification techniques, data representation, and the design of combinational logic networks for decoders, encoders, multiplexers, and demultiplexers. Design of sequential logic devices, including flip-flops, registers, and counters, as well as analysis of devices used to build logic networks, including programmable logic devices. Equivalent to COSC 351.		

CCEE 221L	Logic Design Lab	1(0,2)
This Lab covers design techniques and implementation of combinational and sequential logic circuits. Experiments include logic gates; the design and implementation of logic circuits; combinational logic circuits (decoders, encoders, multiplexers, demultiplexers, and adders); and the design of sequential logic devices using flip-flops, registers, and counters. **Prerequisite**: CCEE 221. **Prerequisite**: CCEE 221. Equivalent to COSC 351L.		

CCEE 331	Signals and Systems	3(3,0)
This course covers the mathematical description and classification of continuous and discrete signals and systems. Topics include types of signals and systems, Fourier series, Fourier transforms, Discrete-Time Fourier transforms (DTFT), Discrete and Fast Fourier Transforms (DFT and FFT), Laplace transforms, z-transforms, and transfer functions. **Prerequisite:** MATH 216		

CCEE 341	Communication Systems	3(3,0)
This course introduces the fundamentals of transmission and reception in communication systems, as well as the effects of noise. Topics include power spectral density; amplitude modulation and demodulation; angle modulation and demodulation; analog communication system performance in the presence of noise; sampling and analog-to-digital conversion; introduction to digital modulation; and channel capacity. **Prerequisite**: CCEE 331.		

CCEE 426	Design of Embedded Systems	3(3,0)
This course addresses the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, operating systems and scheduling, system-level simulation, and interface-based design. **Prerequisite**: CCEE 221. Equivalent to COSC 356.		

CCEE 426L	Design of Embedded Systems Lab	1(0,2)
This lab complements topics covered in the CCEE 426 course. It involves the design of embedded real-time systems, models of computation, validation techniques, and automatic synthesis. Experiments include design using finite-state machines, synchronous languages, data-flow networks, Petri nets, software optimization and performance estimation, scheduling, system-level simulation, and interface-based design. **Co-requisite**: CCEE 426. Equivalent to COSC 356L.		

CCEE 460	Artificial Intelligence	3(3,0)
This course introduces students to the fundamentals of knowledge representation and learning methods in artificial intelligence. The emphasis will be on understanding fundamental concepts in artificial intelligence and applying corresponding approaches to solve practical problems and develop useful software applications. Covered topics include intelligent agents, informed and uninformed search strategies, and adversarial search. **Prerequisite**: CCEE 214 or equivalent.		

CCEE 460L	Artificial Intelligence Lab	1(0,2)
This lab will help prepare students to handle the huge amount of data produced by both humans and machines today. The student will learn to use state-of-the-art AI tools to analyze data and make complex decisions based on that data using real-world practical examples and case studies. At the end of the lab, the student will be able to apply AI knowledge in their field at work or in day-to-day life to make better decisions. **Co-req.:** CCEE 460.		

CCEE 534	Signal Processing	3(3,0)
This course covers time series analysis, frequency analysis, time-frequency, and time-scale analysis. It also covers the design of digital filters and signal modeling. **Prerequisite**: CCEE 331.		

ENGL 210	English Composition and Rhetoric	3(3,0)
This course reviews the fundamentals of good academic writing in English, teaches essay writing essentials and research skills in two rhetorical modes (persuasion and argumentation), and provides practice in writing essays in these modes, a research paper, and an oral presentation. **Prerequisite:** ENGL 101 or TOEFL 550+ (paper) or 80+ (computer).		

ENGL 217	Professional English Communication	3(3,0)
This required course is designed to help students develop effective professional communication skills, both orally and in writing. In this course, students learn how to write emails, memos, letters, proposals, reports, and other forms of employment correspondence. In addition, this course helps students sharpen their presentation skills. Broadly, this course enables students to behave professionally and effectively in their prospective jobs. **Prerequisite**: ENGL 210.		

ENGR 300	Engineering Economics and Management	3(3,0)
Concepts and techniques in basic Engineering economy principles and applications. Interest and financial mathematics; present worth, annual worth, benefit/cost ratio, internal rate of return, multiple alternatives, income tax, inflation, Risk analysis, Investment and investment choice, equivalence, loans, cost of capital, retirement and replacement, sector analysis and viewpoint, sensitivity analysis, accounting, and financial statements.		

ENGR 510	Engineering Project Management	3(3,0)
The course covers key components of engineering project management including projects election and planning, project time management, cost estimation and pricing, contract and specifications, quality management, engineering ethics and professional conduct, realizing impact of engineering solutions in various contexts (global, economic, environmental, societal, etc...), sustainability in engineering designs, human resources consideration, communications, risk management, and procurement management. **Prerequisite:** ENGR 300.		

MATH 215	Calculus III	3(3,0)
This course covers complex numbers, trigonometric functions, inverse trigonometric functions, hyperbolic functions, and integration techniques. The course also covers infinite sequences and series: limits of sequences of numbers, bounded sequences, integral test for series, comparison tests, ratio and root tests, alternating series, absolute and conditional convergence, power series, Taylor and MacLaurin series, and applications of power series. Polar functions, polar coordinates, and graphing of polar curves are also covered. In addition, topics from multivariable calculus are introduced: double integrals, applications to double integrals, and double integrals in polar form.		

MATH 216	Calculus IV	3(3,0)
Multivariable calculus and vector analysis for three-dimensional problems. Topics include 3-D vectors, dot and cross products; quadratic surfaces; limits, continuity, and partial derivatives; gradient, chain rule, and directional derivatives; tangent planes, differentials, extrema, and saddle points; triple integrals in rectangular, cylindrical, and spherical coordinates with applications to mass and moments; vector fields, work, flux, and Green's Theorem; and an introduction to Fourier series. **Prerequisite:** MATH 215.		

MATH 311	Linear Algebra with Applications	3(3,0)
Systems of linear equations, matrix algebra, linear transformations, determinants, vector spaces, eigenvalues and eigenvectors, symmetric matrices, orthogonality, diagonalization.		

MATH 314	Ordinary Differential Equations	3(3,0)
First-order linear differential equations; linear differential equations of second and higher order; linear differential equations with variable coefficients; series solutions; systems of differential equations; Laplace transforms and their inverses. **Prerequisite:** MATH 216.		

MATH 351	Probability and Statistics	3(3,0)
Probability and conditional probability; discrete and continuous random variables; marginal distributions; expectation, variance, mean, median, covariance, and correlation; conditional expectation; Normal distribution; sampling distribution; prediction and confidence intervals; hypothesis testing; and regression line and correlation coefficients. **Prerequisite**: MATH 215		

MATH 421	Numerical Analysis	3(3,0)
Error Analysis, solutions of nonlinear equations using fixed-point, Newton-Raphson-Muller's methods, solution of linear systems using Gaussian elimination-iterative methods, interpolation and approximation using Taylor series-Lagrange approximation-Newton polynomials, numerical differentiation and integration, numerical optimization, solutions of ordinary and partial differential equations using Euler's and Heun's and Rung-Kutta methods, MATLAB codes related to the above-mentioned topics. **Prerequisite**: MATH 311.		

MECH 333	Thermal Fluid Sciences	3(3,0)
This course seeks to impart thermal-fluid fundamental concepts (including Thermodynamics, Fluid Mechanics, and Heat Transfer) to non-mechanical engineering students. These cover: thermodynamic state and properties of pure substances; concepts of work and heat; energy analysis of closed systems (First Law) and simple flow devices; fluid statics; Bernoulli's equation; Conservation of mass, momentum & energy; viscous flow in pipes; introduction to conduction, convection and radiation heat transfer. **Prerequisite**: MATH 216. Annually.		

PHYS 211	Physics: Electricity and Magnetism and Lab	3(3,1)
This course covers fundamental topics in Electricity and Magnetism: Electric forces and Electric Fields for discrete and continuous charge distributions; Gauss's Law; Electric Potential; Capacitance and Dielectrics; Kirchhoff's rules; Magnetic Fields and Forces; Biot-Savart Law; Ampere's Law; Magnetic Flux and Gauss's Law in Magnetism. It also includes a laboratory component that introduces students to the "real world" through experiments in electricity and magnetism. **Prerequisite**: None.		

PHYS 312	Modern Physics	3(3,0)
This course introduces the principles of revolutionary developments of the 20th century. It covers the interaction of light and matter (Photoelectric effect, Compton effect, Auger effect, etc.), the dual nature of light, various models of atomic description, quantum numbers, the relativistic approach, the Heisenberg Uncertainty Principle, the Schrödinger Equation, and an introduction to the band theory of solids and to particle physics.		

Major Courses

Mandatory electrical engineering courses are described below.

ELEC 200	Introduction to Electrical and Computer Engineering	3(3,0)
In this course, the student will be trained in core skills to excel in electrical, biomedical, computer, and communications engineering. The student will be trained in application programming using an appropriate programming environment. Students will build on the basics of Programming learned in the introductory programming course to implement Digital Signal Processing applications and perform data analysis. Students will also be introduced to basic lab equipment such as Oscilloscopes, Function Generators, and Multi-Meters. Students will be introduced to microcontroller-based projects to explore more of the Electrical and Computer Engineering majors (CCE, BIOM, and ELEC majors). Throughout the course, students will work in multidisciplinary teams to study engineering cases for the community and implement a project to address a community need.		

ELEC 210	Electric Circuits	3(3,0)
This course covers circuit elements and laws; mesh and node equations; network theorems; energy-storage elements; RC, RL, and RLC circuits; transformers;		

sinusoidal (AC) steady-state analysis; power calculation; and an introduction to three-phase circuits. **Prerequisite**: PHYS 191 or Equivalent.

ELEC 301	**Sensors and Sensor Circuit Design**	**3(3,0)**
This course provides an introductory overview of the multidisciplinary field of sensors. It includes interpreting relevant signals; measuring noise, pressure, flow, motion, and force; and electrode theory. **Prerequisite**: ELEC 210		

ELEC 301L	**Sensors and Sensor Circuit Design Lab**	**1(0,2)**
The lab teaches students how to measure and interpret signals. Experiments include signal, noise, pressure (strain gauge), and temperature measurements; fluid and gas flow, motion, and force measurements; and applications of electrodes in medical equipment and in research and simulation kits—co-requisite: ELEC 301.		

ELEC 320	**Electronics**	**3(3,0)**
This course covers semiconductors, PN junctions, diode theory and circuits, bipolar junction transistors, transistor fundamentals, transistor biasing, amplifiers, MOSFETs, and operational amplifiers. **Prerequisite:** ELEC 210.		

ELEC 320L	**Electronics Lab**	**1(0,2)**
This is a Lab course with experiments in Electrical and Electronic Circuits. It covers passive electrical elements and sources; lab instruments; voltage divider circuits; Thevenin's theorem; RC circuits; diode circuits; Op-Amp circuits; BJT and MOSFET characteristics. **Co-requisite:** ELEC 320.		

ELEC 323	**Electronics Circuit Design**	**3(3,0)**
This course offers students a methodology for the design process with an emphasis on the feasibility and preliminary design phases, including conditioning circuits, Op-amps and feedback systems, filters, oscillators, sample/hold, DAC/ADC, and data acquisition Boards. Students will learn how to write proposals and reports and increase their creativity through industry-based group projects on interdisciplinary topics. **Prerequisite**: ELEC 301; ELEC 320.		

ELEC 330	**Power Systems**	**3(3,0)**
This course covers three-phase power systems, symmetrical components, transmission lines, power transformers, power system modeling, per-unit analysis, power flow studies, and fault analysis. **Prerequisite:** ELEC 210.		

ELEC 331	**Electric Machines**	**3(3,0)**
This course covers magnetic circuits; fundamentals of electromechanical energy conversion; and the construction, testing, steady-state analysis, and operational characteristics of DC machines, transformers, induction machines, and synchronous machines. **Prerequisite**: ELEC 210.		

ELEC 331L	Electric Machines Lab	1(0,2)
This is a Lab course with experiments on electric machines. It covers the following topics: the basics of DC motors and generators; DC series, shunt, and separately excited motors and generators; single-phase AC induction machines; and 3-phase machines. **Co-requisite:** ELEC 331.		

ELEC 340	Engineering Electromagnetics	3(3,0)
This course covers vector analysis, static electric fields, static magnetic fields, time-varying fields, Maxwell's equations, and electromagnetic waves. **Prerequisite:** PHYS 211.		

ELEC 400	Summative Learning (BS) Project	3(3,0)
Team-oriented, project-based experience culminating in the creation of an artifact; milestones include project selection and proposal, a creative solution, a report, a presentation, and a demonstration of the device. **Prerequisite**: ENGL 217 taken in the last semester of the BS Program. Annually.		

ELEC 405	Electrical System Design	3(3,0)
This course aims to teach students how to design various electrical systems, including equipment selection, protection systems, EMC and safety requirements, international standards, drawing and interpreting schematics and wiring diagrams, cost considerations, and energy efficiency. **Prerequisite:** ELEC 210.		

ELEC 406	Modeling and Simulation	3(3,0)
This course aims to teach students how to represent, characterize, understand, analyze, and solve real-world problems through modeling and simulation. In this course, students are introduced to the tools and techniques for modeling and simulating systems, ranging from basic circuits to more complex technical systems across various engineering domains. **Prerequisite:** PHYS 211 and ELEC 210.		

ELEC 431	Introduction to Renewable Energy Systems	3(3,0)
This course aims to provide students with information on renewable energy sources, including solar, wind, and wave energy, as well as biomass energy. This course will also illustrate how to link these sources to the grid and synchronize them to deliver electricity reliably and efficiently. **Prerequisite:** ELEC 210.		

ELEC 432	Power Electronics	3(3,0)
This course familiarizes the students with basic power switch technology and associated electronic circuits. In this course, power electronic circuits and switching devices are studied. Their applications in AC/DC, DC/DC, DC/AC, and AC/AC converters, as well as switching power supplies, are studied. **Prerequisite:** ELEC 320.		

ELEC 432L	Power Electronics Lab	1(0,2)
This Lab course is designed to teach students the fundamentals of power electronics. It covers the following topics: uncontrolled single- and 3-phase rectifiers using diodes;		